EXPERIMENT: Write the prolog program for Medical Diagnosis

Aim

To develop a Prolog program for Medical Diagnosis that identifies a possible disease based on the symptoms provided by the user using facts and logical rules.

Objectives

1. To understand the use of facts, rules, and predicates in Prolog.
2. To develop a simple knowledge-based medical diagnosis system.
3. To identify possible diseases from a given set of symptoms.
4. To understand how Prolog performs logical reasoning and pattern matching.

Algorithm

1. Start the program.
2. Define the diseases and their corresponding symptoms using Prolog rules.
3. Define a predicate symptom/1 to accept symptoms from the user.
4. Ask the user whether they have each required symptom.
5. Match the user's answers with the disease rules.
6. If all required symptoms for a disease are satisfied, display the possible disease.
7. If no disease rule matches, display that the disease could not be identified.
8. Stop the program.

CODE

```
print("===================================")
print("    MEDICAL DIAGNOSIS SYSTEM")
print("===================================")
fever = input("Do you have fever? (yes/no): ").lower()
cough = input("Do you have cough? (yes/no): ").lower()
body_pain = input("Do you have body pain? (yes/no): ").lower()
sneezing = input("Do you have sneezing? (yes/no): ").lower()
```

```python
runny_nose = input("Do you have runny nose? (yes/no): ").lower()
chills = input("Do you have chills? (yes/no): ").lower()
headache = input("Do you have headache? (yes/no): ").lower()
stomach_pain = input("Do you have stomach pain? (yes/no): ").lower()
loss_of_smell = input("Do you have loss of smell? (yes/no): ").lower()
print("\n---------- DIAGNOSIS ----------")
if fever == "yes" and cough == "yes" and body_pain == "yes":
    print("Possible Disease: Flu")
elif cough == "yes" and sneezing == "yes" and runny_nose == "yes":
    print("Possible Disease: Common Cold")
elif fever == "yes" and chills == "yes" and headache == "yes":
    print("Possible Disease: Malaria")
elif fever == "yes" and headache == "yes" and stomach_pain == "yes":
    print("Possible Disease: Typhoid")
elif fever == "yes" and cough == "yes" and loss_of_smell == "yes":
    print("Possible Disease: COVID-19")
else:
    print("Disease not identified.")
    print("Please consult a qualified medical professional.")
```

OUTPUT

```
IDLE Shell 3.12.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug  6 2024, 20:45:27) [MSC v.1940 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shaik arshad basha\AppData\Local\Programs\Python\Python312\x
dc.py
================================
     MEDICAL DIAGNOSIS SYSTEM
================================
Do you have fever? (yes/no): yes
Do you have cough? (yes/no): yes
Do you have body pain? (yes/no): yes
Do you have sneezing? (yes/no): no
Do you have runny nose? (yes/no): no
Do you have chills? (yes/no): no
Do you have headache? (yes/no): no
Do you have stomach pain? (yes/no): no
Do you have loss of smell? (yes/no): no

---------- DIAGNOSIS ----------
Possible Disease: Flu
>>>
                                                              Ln: 20  Col: 0
```

Result

The Prolog program was successfully implemented to diagnose a possible disease based on the symptoms entered by the user.

Conclusion

The Medical Diagnosis system demonstrates how Prolog's facts, rules, and logical inference can be used to solve a simple expert-system problem. The system matches patient symptoms with predefined disease rules and produces a possible diagnosis.